

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

31,623
held-out plate appearances

3.05%
held-out HR rate

0.6020
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-29, scored on the 31,623 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5862	1.61×	0.02950	0.13544	0.00535
KNN comps Prior is the pooled rate of the nearest comparables.	0.6020	1.55×	0.02945	0.13442	0.00478
SVR Support-vector regression on the same feature matrix.	0.5963	1.55×	0.02949	0.13515	0.00527
Random forest Forest regression; a learned similarity metric.	0.5956	1.47×	0.02948	0.13481	0.00643
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6020	1.55×	0.02945	0.13442	0.00478
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6020	1.55×	0.02945	0.13442	0.00478
Linear regression Ordinary least squares on the same features.	0.5963	1.51×	0.02947	0.13486	0.00416
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5974	1.55×	0.02949	0.13494	0.00556
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.53×	0.02948	0.13486	0.00505
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5962	1.55×	0.02949	0.13495	0.00593

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 252 knn 252 svr 252 rf 252 core 252 ensemble 252
linear 252 logistic 252 nn 252 xgb 252

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,518 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5671	1.39×	0.02386	0.11499	0.01167
KNN comps Prior is the pooled rate of the nearest comparables.	0.5821	1.39×	0.02386	0.11479	0.01119
SVR Support-vector regression on the same feature matrix.	0.5696	1.28×	0.02403	0.11926	0.01804
Random forest Forest regression; a learned similarity metric.	0.5845	1.16×	0.02400	0.11611	0.01452
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5821	1.39×	0.02386	0.11479	0.01119
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5821	1.39×	0.02386	0.11479	0.01119
Linear regression Ordinary least squares on the same features.	0.5744	1.05×	0.02396	0.11712	0.01538
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5796	1.28×	0.02395	0.11633	0.01209
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5761	1.39×	0.02393	0.11629	0.01479
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5798	1.16×	0.02403	0.11727	0.01631

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Munetaka Murakami 17.8%, Miguel Vargas 15.3%, Randal Grichuk 14.8%, Kyle Teel 13.2%, Jo Adell 13.0%, Colson Montgomery 11.1%
Flat league	5/6	Munetaka Murakami 16.1%, Miguel Vargas 13.8%, Randal Grichuk 13.4%, Kyle Teel 13.1%, Jo Adell 12.3%, Chase DeLauter 10.8%
KNN comps	6/6	Munetaka Murakami 17.7%, Miguel Vargas 15.1%, Randal Grichuk 14.6%, Kyle Teel 13.7%, Jo Adell 12.8%, Colson Montgomery 11.2%
SVR	6/6	Munetaka Murakami 18.2%, Randal Grichuk 15.4%, Miguel Vargas 15.2%, Jo Adell 13.2%, Kyle Teel 12.7%, Colson Montgomery 10.7%
Random forest	6/6	Munetaka Murakami 18.0%, Miguel Vargas 14.9%, Randal Grichuk 14.4%, Kyle Teel 13.5%, Jo Adell 12.9%, Colson Montgomery 11.3%
Core (KNN+SVR)	6/6	Munetaka Murakami 17.7%, Miguel Vargas 15.1%, Randal Grichuk 14.6%, Kyle Teel 13.7%, Jo Adell 12.8%, Colson Montgomery 11.2%
Forest blend	6/6	Munetaka Murakami 17.7%, Miguel Vargas 15.1%, Randal Grichuk 14.6%, Kyle Teel 13.7%, Jo Adell 12.8%, Colson Montgomery 11.2%
Linear regression	6/6	Munetaka Murakami 17.5%, Miguel Vargas 15.1%, Randal Grichuk 14.5%, Kyle Teel 13.4%, Jo Adell 12.9%, Colson Montgomery 11.5%
Logistic regression	6/6	Munetaka Murakami 17.8%, Randal Grichuk 15.4%, Miguel Vargas 15.4%, Jo Adell 13.2%, Kyle Teel 12.5%, Colson Montgomery 12.0%
Neural net (torch)	6/6	Munetaka Murakami 17.8%, Miguel Vargas 15.2%, Randal Grichuk 15.2%, Jo Adell 13.3%, Kyle Teel 12.8%, Colson Montgomery 11.4%
XGBoost	6/6	Munetaka Murakami 18.1%, Miguel Vargas 15.1%, Randal Grichuk 14.6%, Kyle Teel 13.4%, Jo Adell 13.1%, Colson Montgomery 11.4%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
1Munetaka Murakami	slate17.8%	16.1%	17.7%	18.2%	18.0%	17.7%	17.7%	17.5%	17.8%	
2Miguel Vargas	CWS15.3%	13.8%	15.1%	15.2%	14.9%	15.1%	15.1%	15.1%	15.4%	
3Randal Grichuk	CWS14.8%	13.4%	14.6%	15.4%	14.4%	14.6%	14.6%	14.5%	15.4%	
4Kyle Teel	CWS13.2%	13.1%	13.7%	12.7%	13.5%	13.7%	13.7%	13.4%	12.5%	
5Jo Adell	LE13.0%	12.3%	12.8%	13.2%	12.9%	12.8%	12.8%	12.9%	13.2%	
6Colson Montgomery	CWS11.1%	10.1%	11.2%	10.7%	11.3%	11.2%	11.2%	11.5%	12.0%	
7Nathaniel Lowe	CLE10.9%	10.6%	10.9%	10.7%	11.2%	10.9%	10.9%	10.8%	10.7%	
8Chase DeLauter	CLE10.7%	10.8%	10.6%	10.5%	10.6%	10.6%	10.6%	10.5%	10.4%	
9Angel Martínez	CLE10.7%	10.7%	10.5%	10.7%	10.6%	10.5%	10.5%	10.5%	10.5%	
10Travis Bazzana	CLE10.0%	10.0%	9.8%	9.6%	10.0%	9.8%	9.8%	9.8%	9.6%	
11José Ramírez	CLE9.9%	9.9%	9.7%	9.5%	9.8%	9.7%	9.7%	10.0%	10.2%	
12Tristan Peters	CWS9.7%	10.3%	9.9%	10.2%	10.1%	9.9%	9.9%	10.0%	10.1%	
13Sam Antonacci	CWS9.0%	10.8%	9.4%	8.5%	9.5%	9.4%	9.4%	9.2%	9.1%	
14Chase Meidroth	CWS7.9%	9.2%	8.2%	8.6%	8.5%	8.2%	8.2%	8.3%	8.2%	
15Braden Montgomery	CWS7.5%	8.4%	7.2%	7.0%	7.2%	7.2%	7.2%	7.0%	6.9%	
16Bryan Rocchio	CLE6.7%	7.2%	6.5%	6.8%	6.4%	6.5%	6.5%	6.4%	6.4%	
17Austin Hedges	CLE6.1%	7.5%	6.7%	6.6%	6.6%	6.7%	6.7%	6.7%	6.4%	
18Steven Kwan	CLE4.6%	7.8%	5.5%	4.0%	4.0%	5.5%	5.5%	3.3%	4.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.